

The background features a large white circle in the center, partially overlapping a light blue area on the left and a light pink area on the right. A dark blue shape is at the bottom, also overlapping the white circle. The text is centered within the white circle.

DEVELOPER SURVEY DATA ANALYSIS

OVERALL SUMMARY:

- This is an exploratory data analysis on developer survey data.
- Data science questions relating to the jobs and life of the respondents were explored.
- The survey included people from as many as 135 countries with USA dominating with most respondents. The age group of the respondents are relatively young.
- Almost all have jobs and are satisfied with their career choices.
- Majority of the respondents use Stackoverflow for their day-today work.
- This survey provided valuable insight into lives of developers around the world.

BACKGROUND

- The data that we are exploring is a developer survey data aimed at Stack Overflow users.
- Stack Overflow has always been a place for reference for developers and coders who may have come across an obstacle in their system. It is a platform for discussion and helping one another in situations that a developer may have faced earlier.
- This survey has been answered by 11398 people ranging from over 130 countries. The questionnaire included around 70 questions.
- This survey provides a lot of raw data that can provide an insight into the lives of developers. Carefully conducting analysis is must in order to extract the full potential of the data.

DATA SCIENCE QUESTIONS:

Demographics:

- What are the age ranges of the respondents?
- Which countries are the respondents mostly from?
- Do they code as a hobby?
- Do they have family responsibilities?

StackOverflow Use:

- How often do they visit StackOverflow?
- How much time do they spent on it?
- Does StackOverflow really help?

Job Information:

- Are all of them employed?
- Are they satisfied with their jobs?
- How often do they work from home?
- How many work hours do they have per week?
- How much do they make depending on country?

Software used:

- Which are the most used languages?
- Which are the most used databases?
- Which are the most used webframes?
- Which are the most used platforms ?

OBJECTIVES

- To find out and visualize more about the demographics of the respondents.
- To find out and visualize about the jobs of the respondents
- To find out how dependent developers are on StackOverflow
- To visualize the software used by majority respondents.

METHODOLOGY:

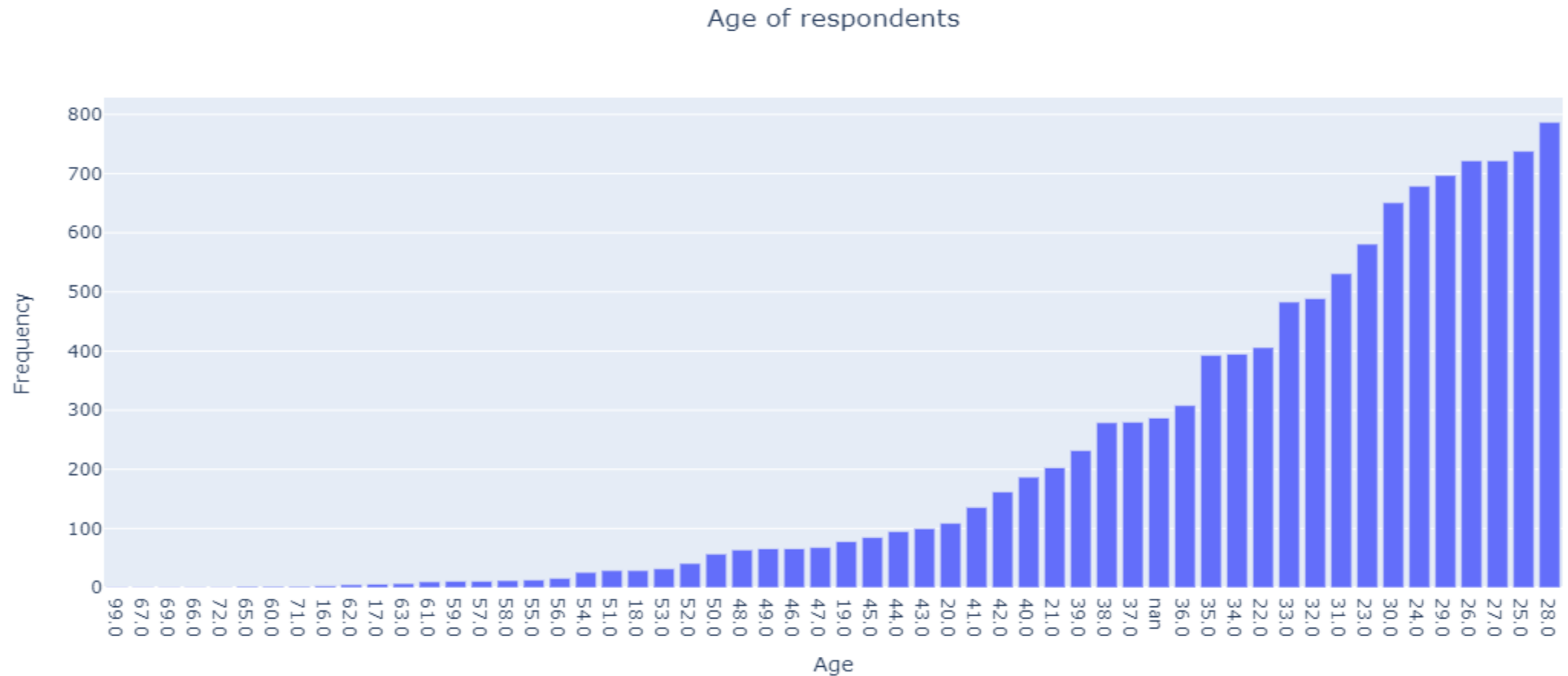
- The main tool for this exploratory data analysis is python.
- Various libraries such as pandas, seaborn, matplotlib, numpy etc were used to make this analysis possible.
- Other online sources were used as reference points to get analysis ideas.

RESULTS & VISUALIZATIONS:

1. DEMOGRAPHIC INFO

7

1. The following graph displays the age of the respondents along with their frequency. It can be seen that most of the respondents are in their 20s and in the early 30s. This shows that most developers are young and upcoming in the industry.

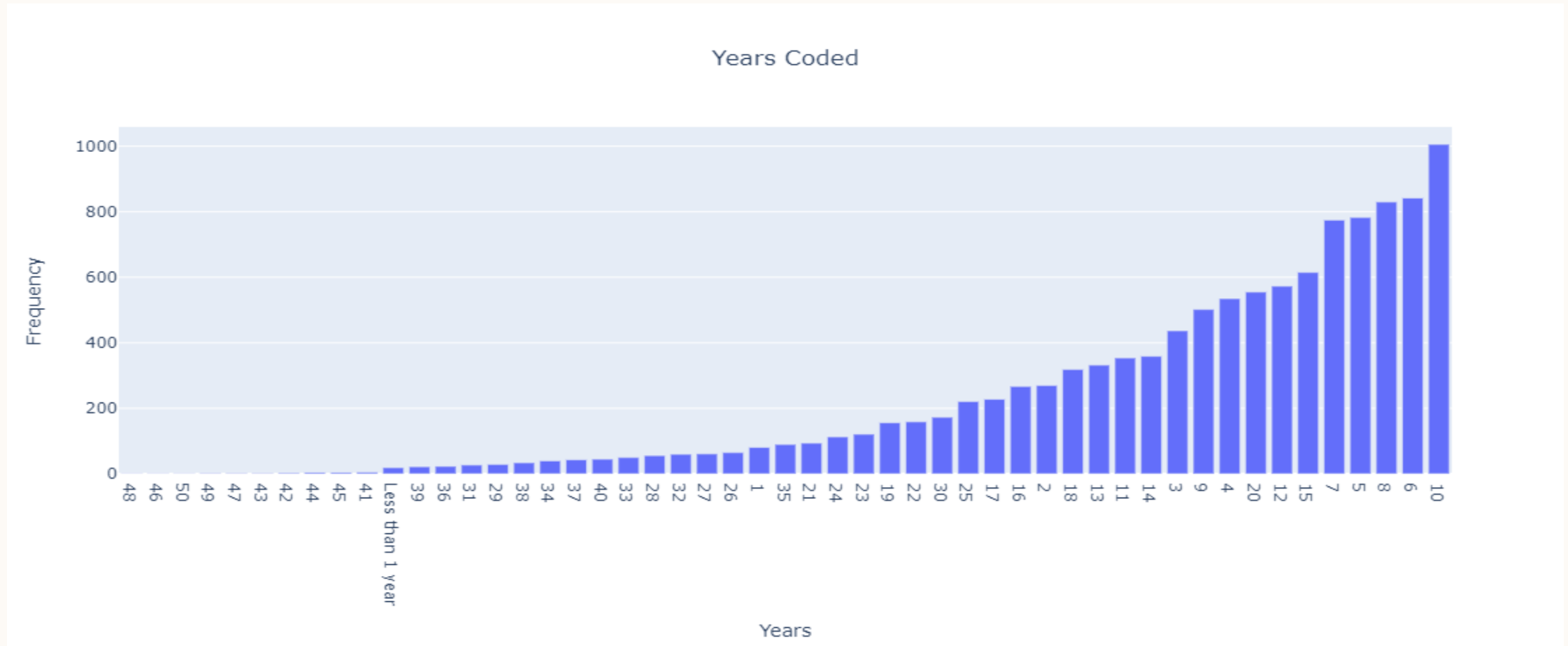


RESULTS & VISUALIZATIONS:

1. DEMOGRAPHIC INFO

8

2. The following graph shows how many years the developers have been coding for. It seems that top ten in average, the developers have coded for around 10 years. This may tell us that the respondents were not beginners when it came to coding.

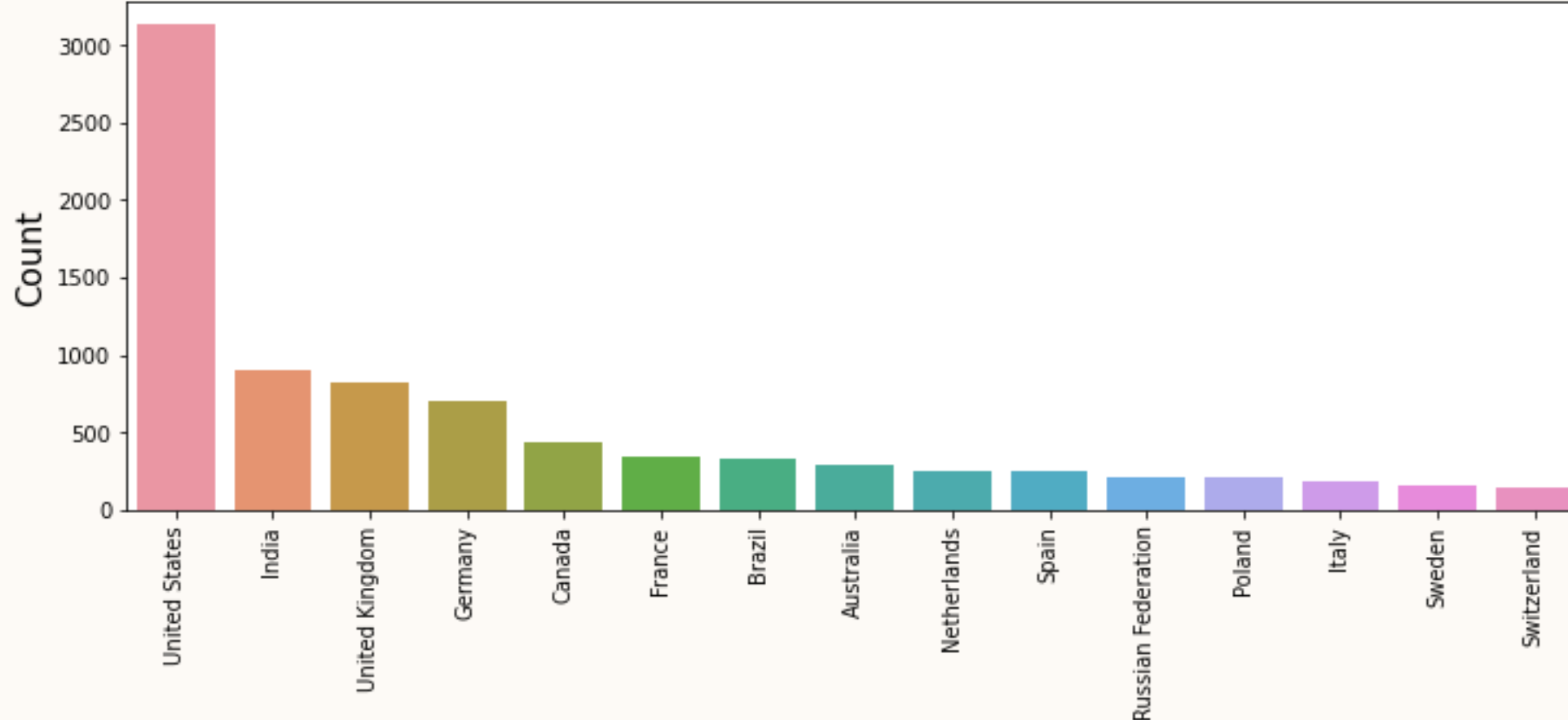


RESULTS & VISUALIZATIONS:

1. DEMOGRAPHIC INFO

3. Next we would like to see where these respondents are from. Since there were more than 135 countries we look at the top 15 countries by number that responded to the survey. It can be seen that the USA has maximum respondents with above 3000 answers. The following 3 countries are almost same in terms of respondents with less than a 1000 answering.

TOP 15 Countrys of the Survey Respondets



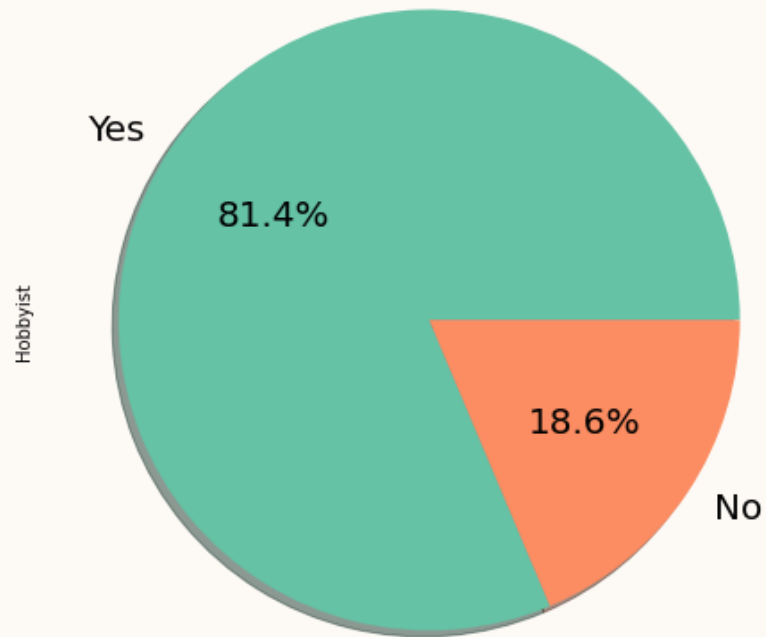
RESULTS & VISUALIZATIONS:

1. DEMOGRAPHIC INFO

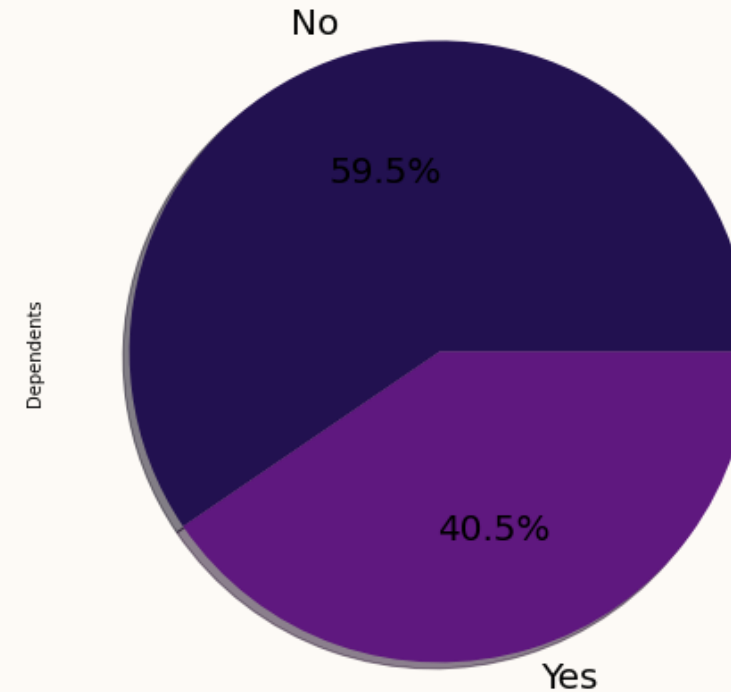
10

4. In an insight into their personal lives, we can see that over 80 percent of them code as a hobby. This also supports the fact that most people have been coding around 10 years and since most of them are young, coding must be a hobby. The second pie graph shows that more than half of them are having dependents to look after.

Percentage of people that code as hobby



Percentages of developers having dependents

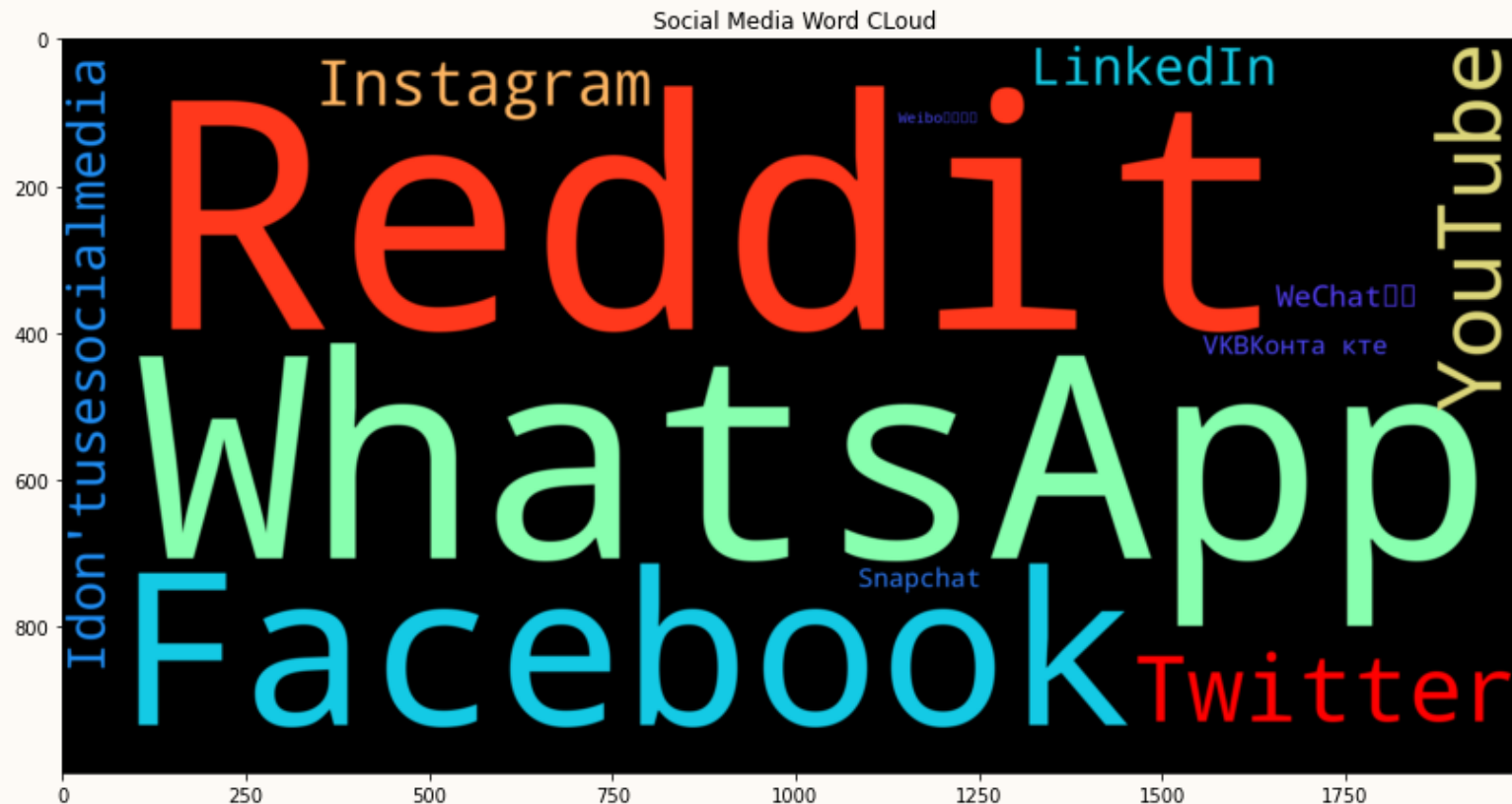


RESULTS & VISUALIZATIONS:

1. DEMOGRAPHIC INFO

11

5. Looking at their social media used, there are expected platforms such as WhatsApp, Facebook and Twitter. However equally popular seems to be the platform Reddit which is fairly uncommon or unpopular among the other social media giants.

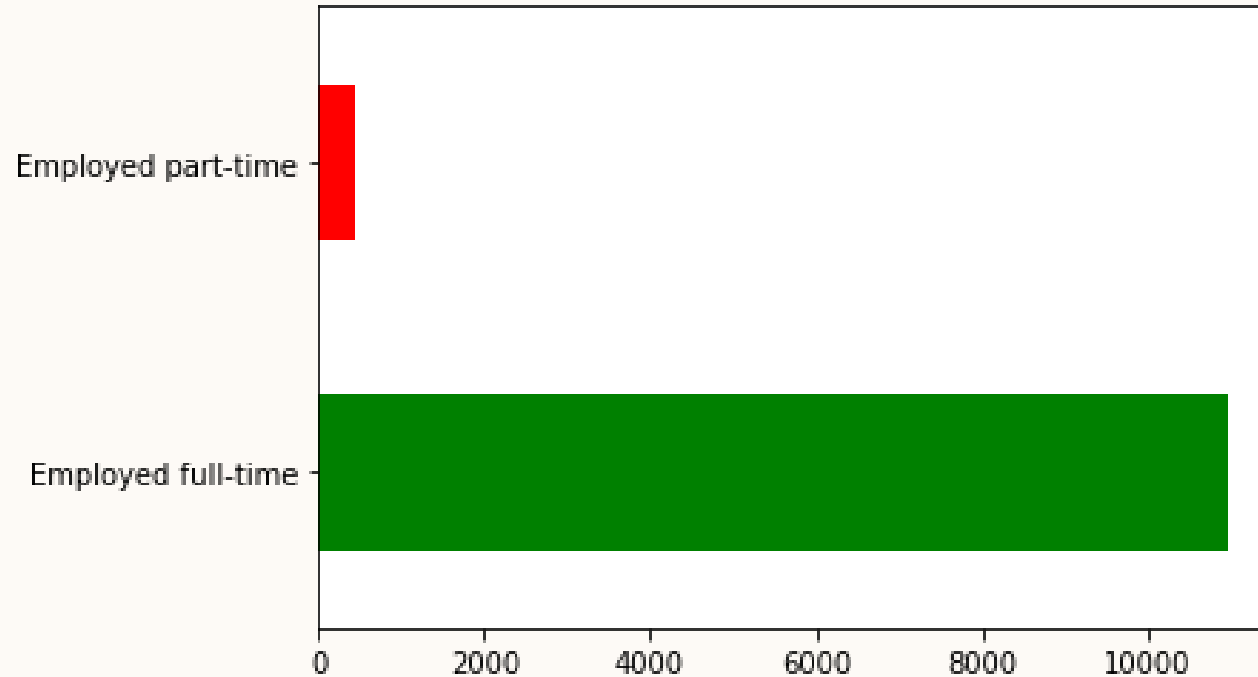


RESULTS & VISUALIZATIONS:

2. JOB RELATED INFO

12

1. From the following bar-graph it is clear that 95 percent of the developers and employed full time. This is good news when 60 percent of them have dependents.

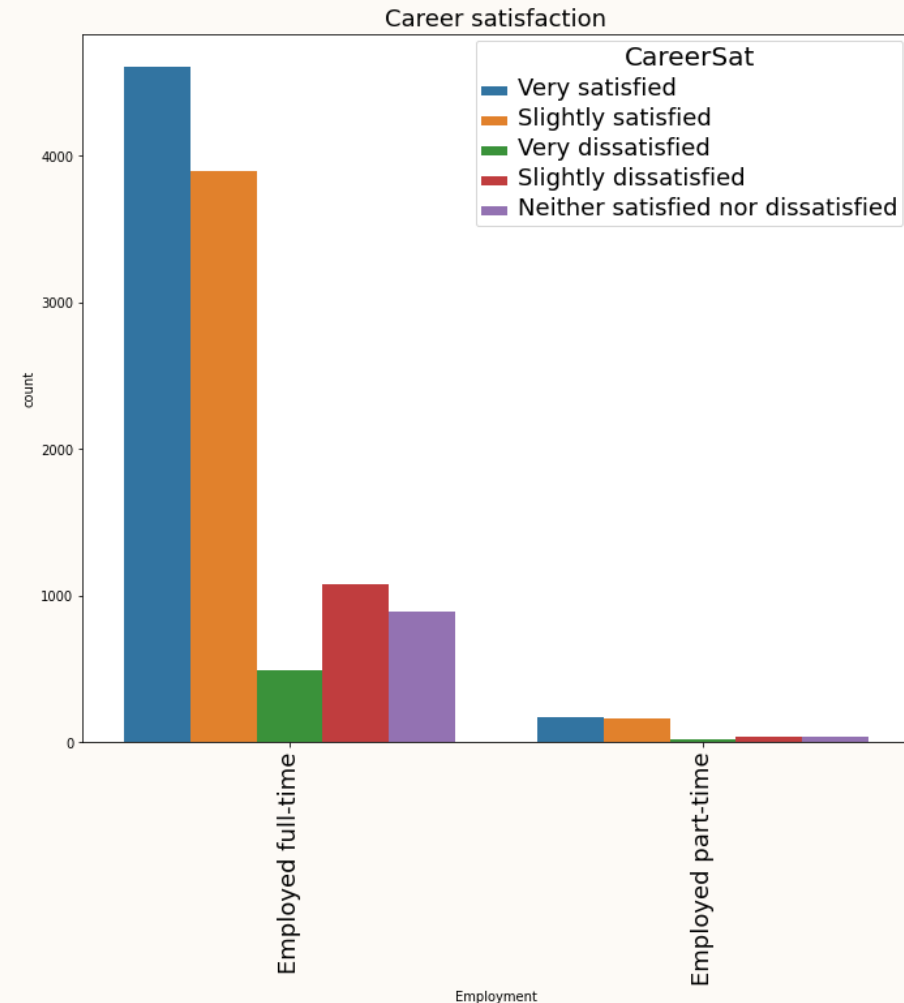
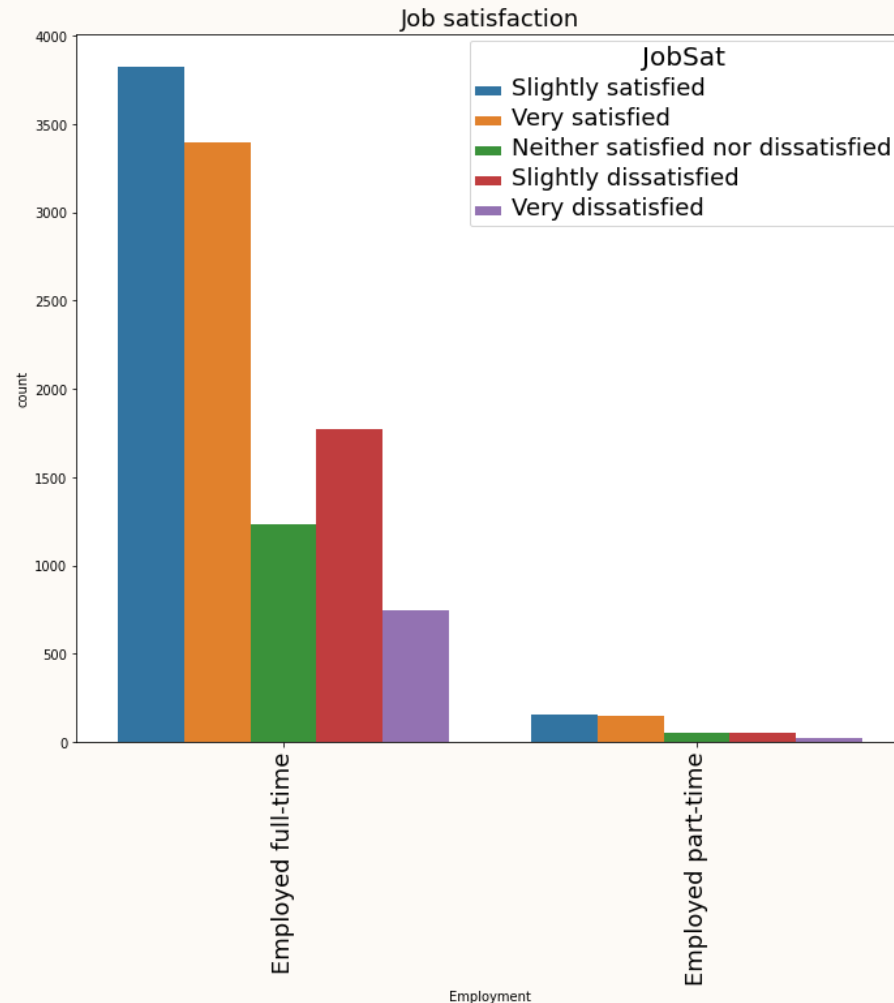


RESULTS & VISUALIZATIONS:

2. JOB RELATED INFO

13

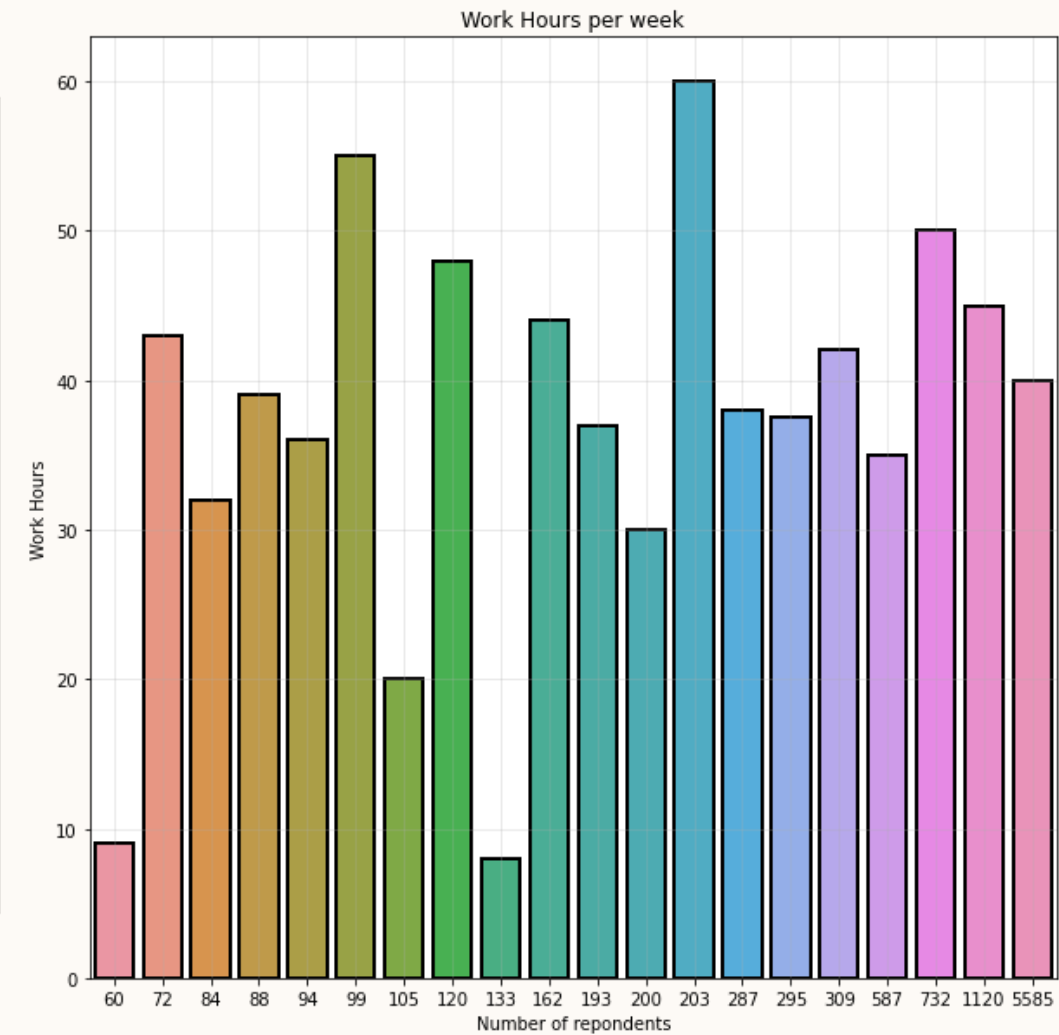
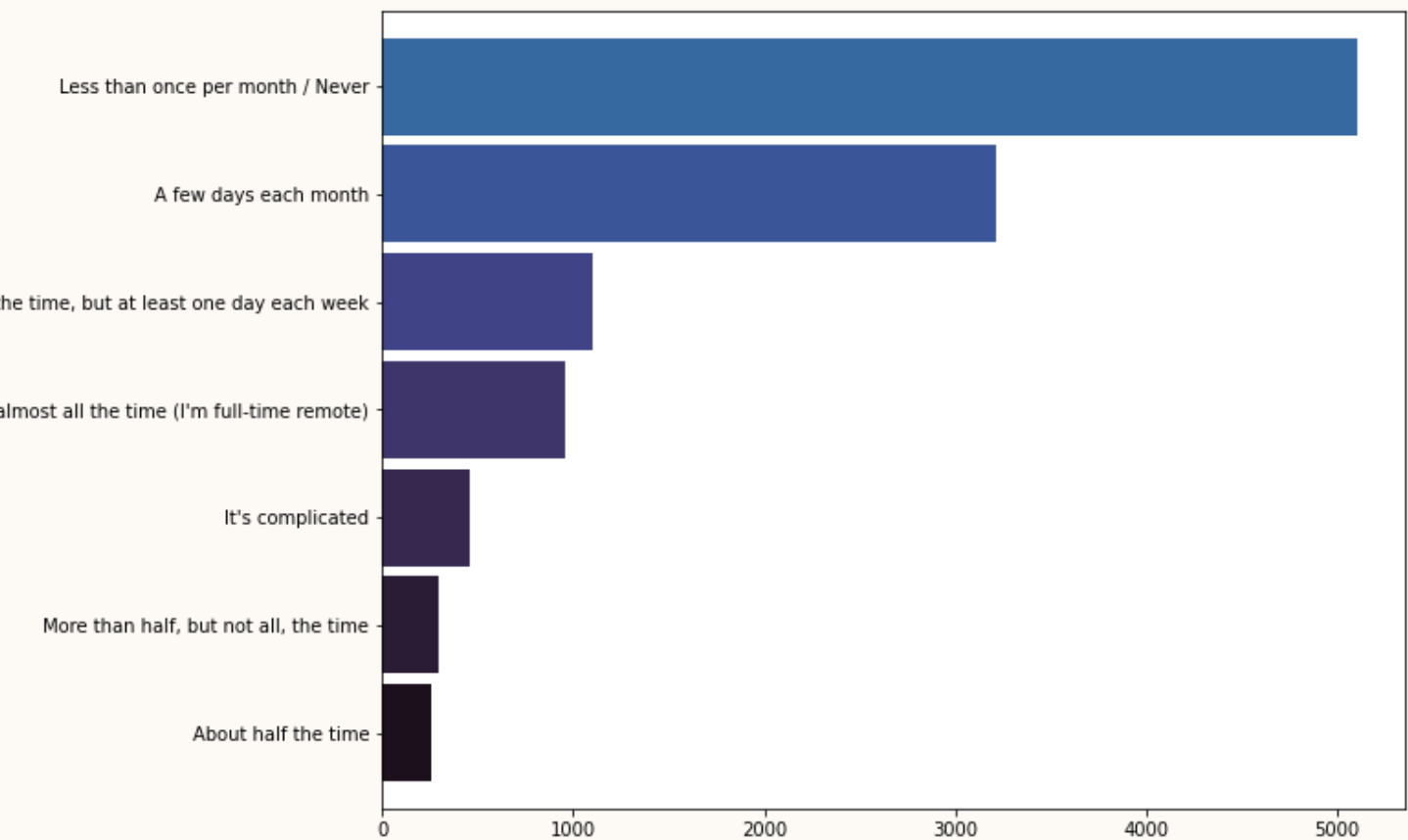
2. The following 2 bar plots provides an insight into their job and career satisfaction. If we compare, we can see that although the majority is slightly satisfied with their jobs, they are very satisfied with their careers.



RESULTS & VISUALIZATIONS:

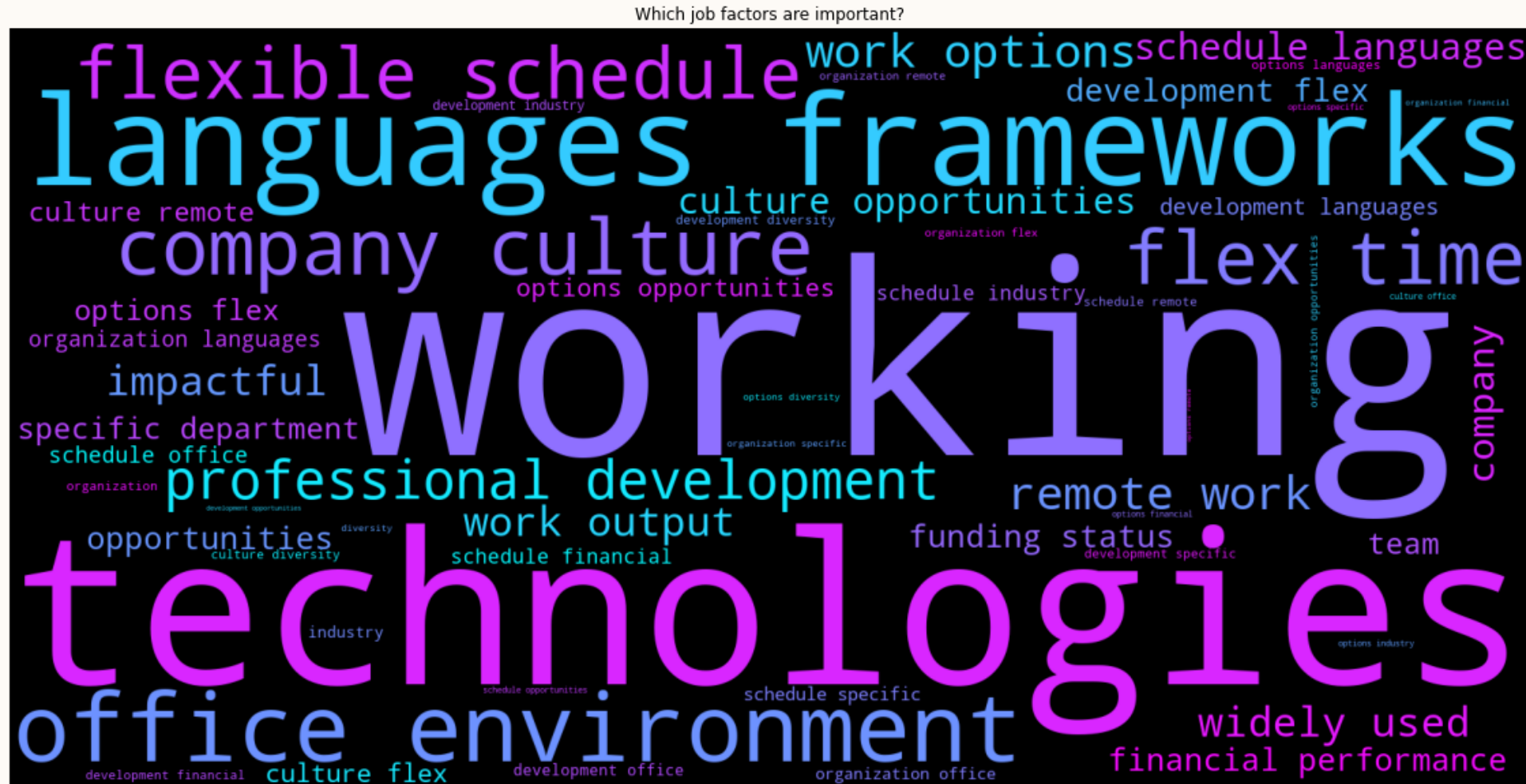
2. JOB RELATED INFO

3. The following to plots tell us how often they work remotely and how long they work. Although work from home can be possible for developer jobs, majority do it less than once a month. The developers also seem to do around 40 hour weeks on average.



RESULTS & VISUALIZATIONS: 2. JOB RELATED INFO

4. This word cloud represents what job factors they believe are important. Among the notable are languages learned, flexible schedule, frameworks & technologies used, and the office environment.



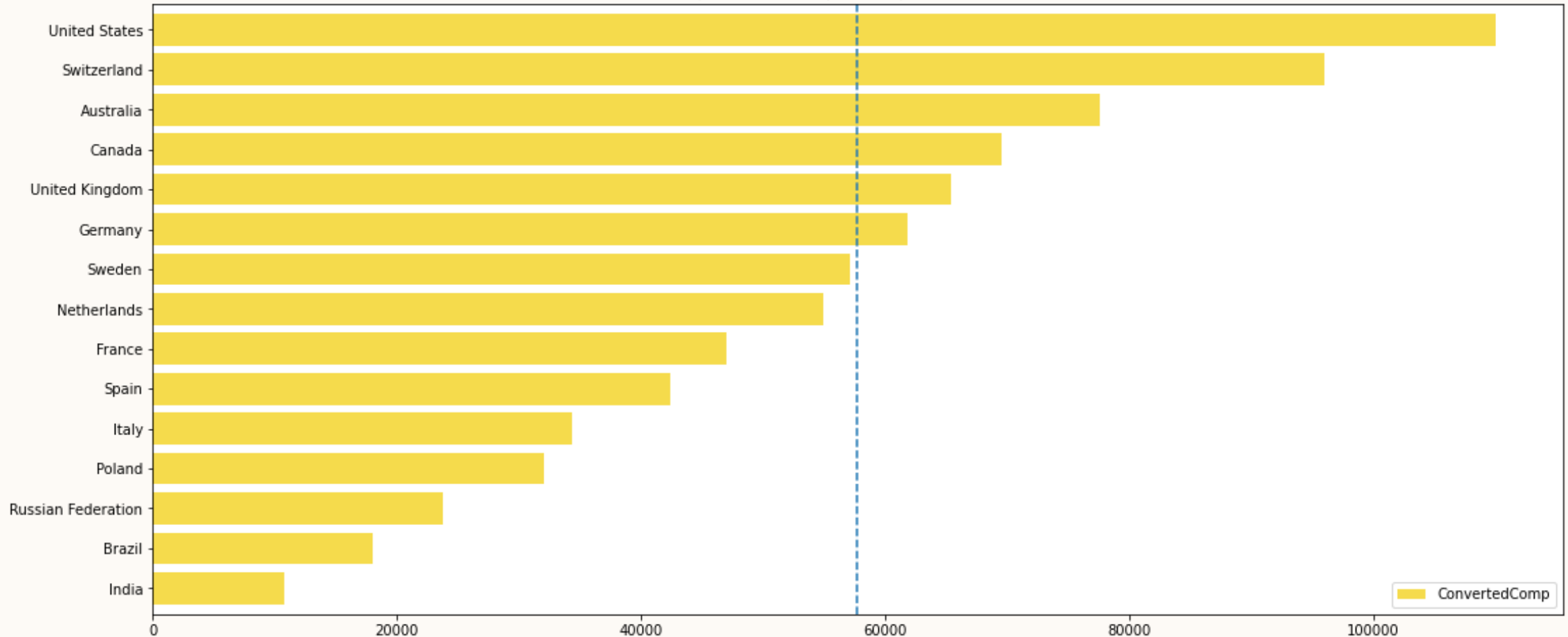
RESULTS & VISUALIZATIONS:

2. JOB RELATED INFO

16

5. This horizontal plot shows us the annual salary compensation depending on country converted to US dollars. The median salary seems to be a bit shy of \$60,000. As expected, USA being a tech hotspot of the world, tops the list with over a 100k annual income. Surprisingly India, which had the second highest respondents has the lowest annual salary .

Compensation of Top 15 Respondent Countries

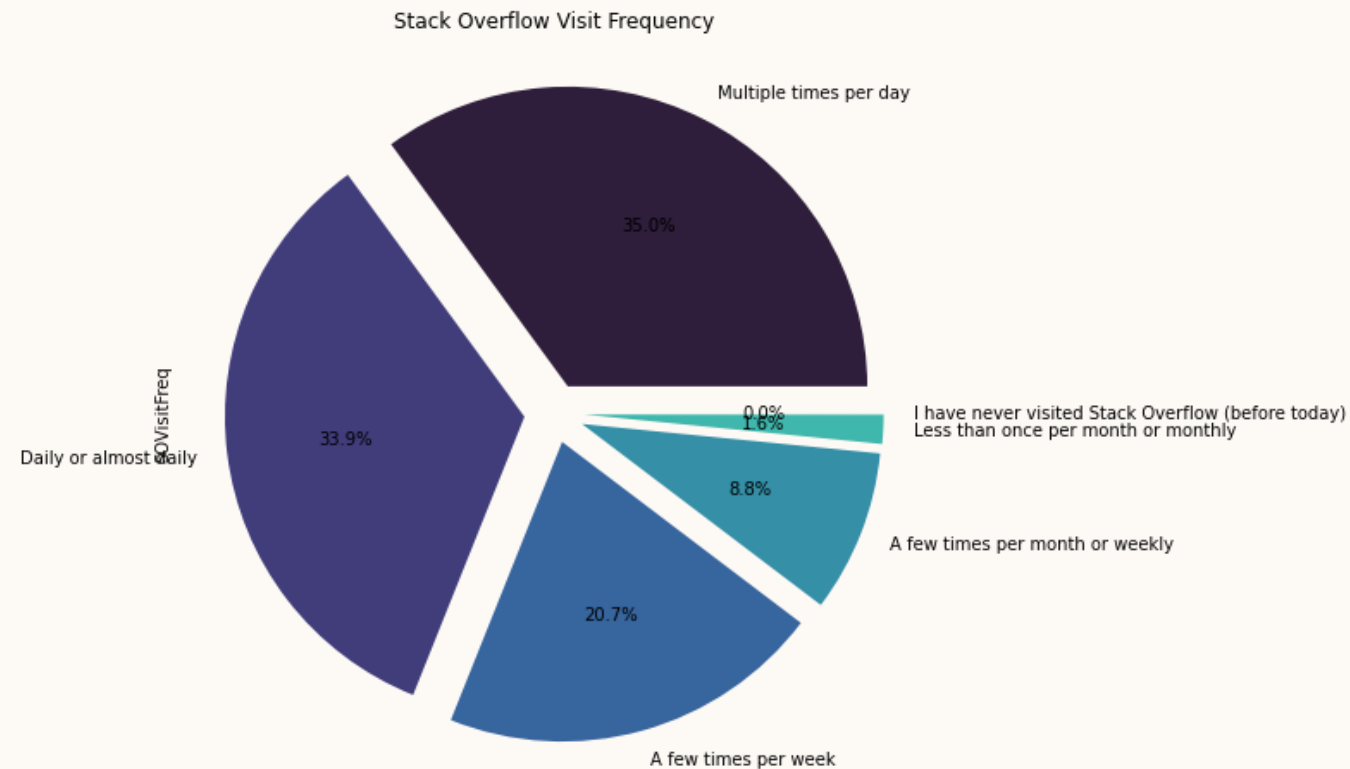
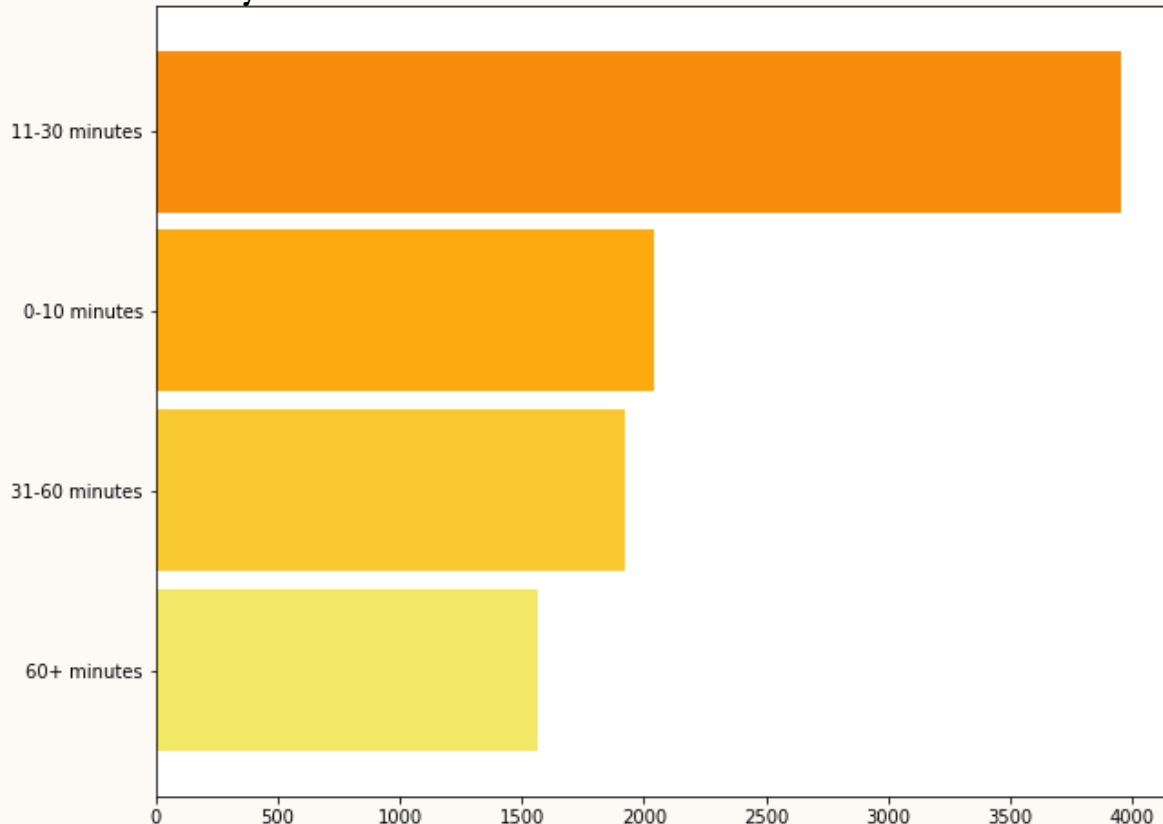


RESULTS & VISUALIZATIONS:

3. STACK OVERFLOW DEPENDENCY

17

1. The following bar plot shows how much developers spent on average in Stack overflow. Majority of them spend half an hour or less. The pie graph displays how many times the site is frequented. This shows that even though they are experienced developers, they still need help as majority frequents the site almost daily.

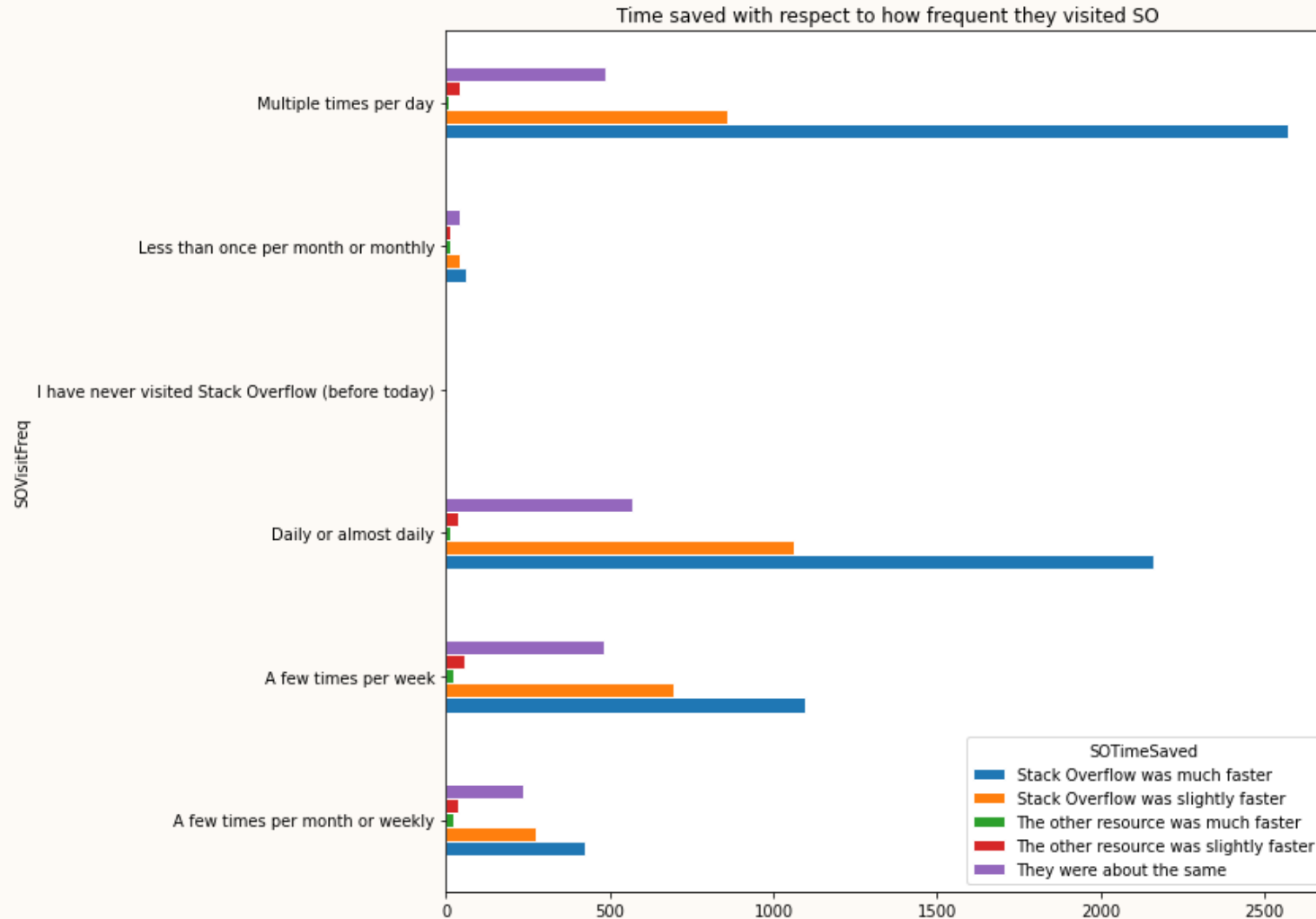


RESULTS & VISUALIZATIONS:

3. STACK OVERFLOW DEPENDENCY

18

2. This plot shows how much time was saved with respect to how frequently they visited. Those that visited frequently claim that Stack Overflow helped them save time by providing answers.

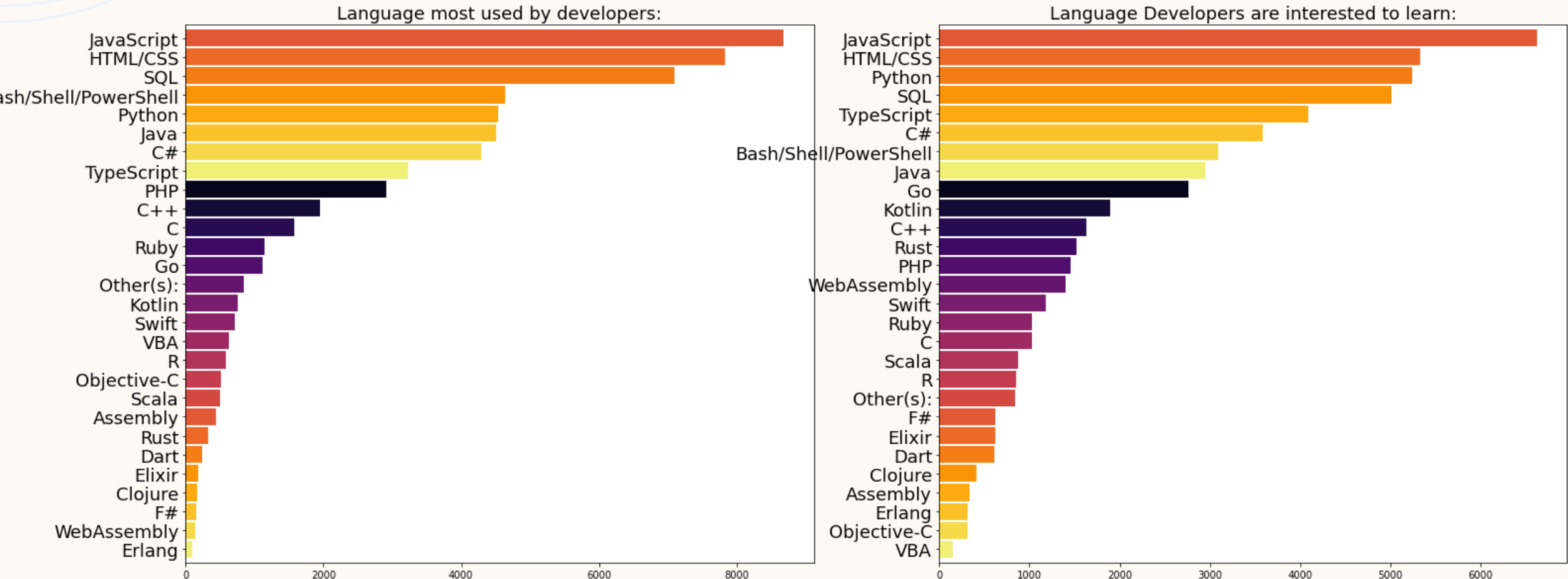


RESULTS & VISUALIZATIONS:

4. SOFTWARE USED

19

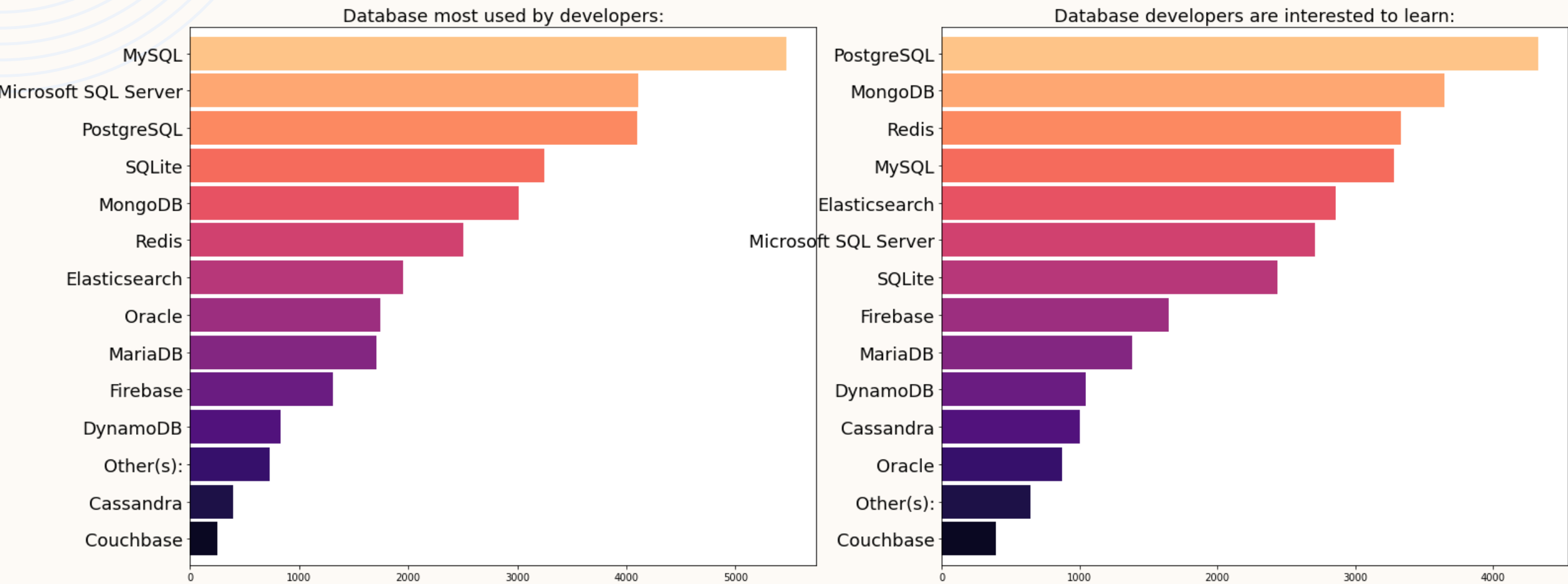
1. Next we look into the software used by the developers. From the following graphs we can see that the most commonly used languages are for web development mainly Javascript and HTML. Interestingly an unpopular language Bash/Shell/Powershell are higher than popular C++ or Java.



RESULTS & VISUALIZATIONS:

4. SOFTWARE USED

2. As expected, MySQL being the popular database, is also the one most used by developers.

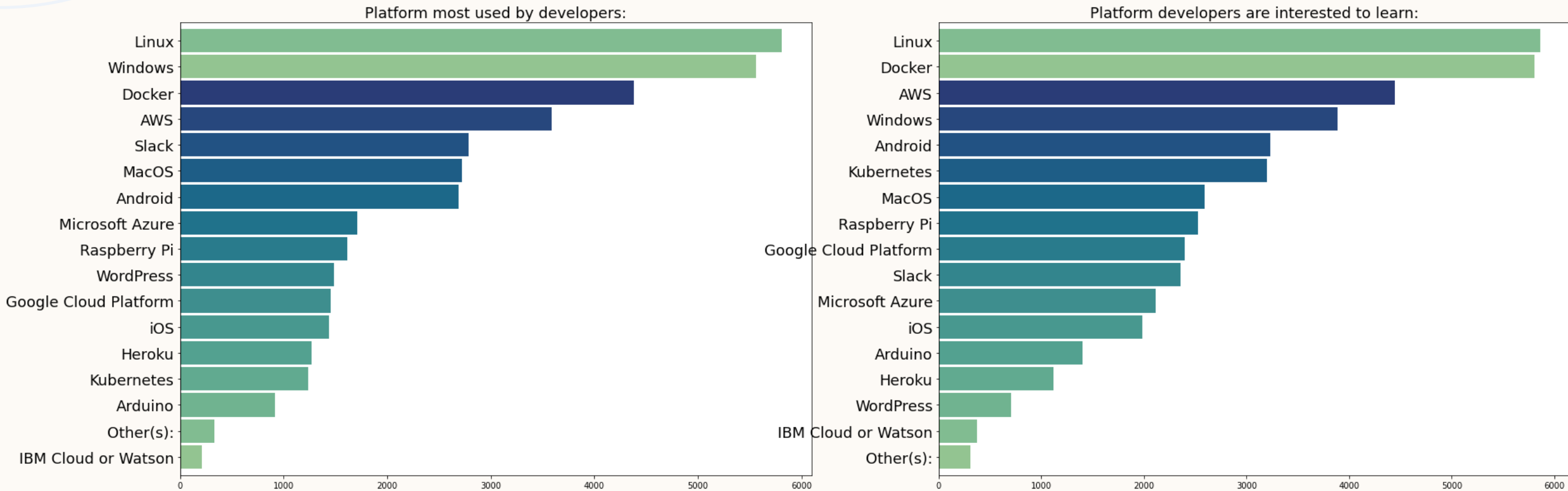


RESULTS & VISUALIZATIONS:

4. SOFTWARE USED

21

4. The most used platform surprisingly is linux with the expected platform Windows in a close second. There is however an overwhelming amount of respondents that are interested in learning to use Linux and Docker.

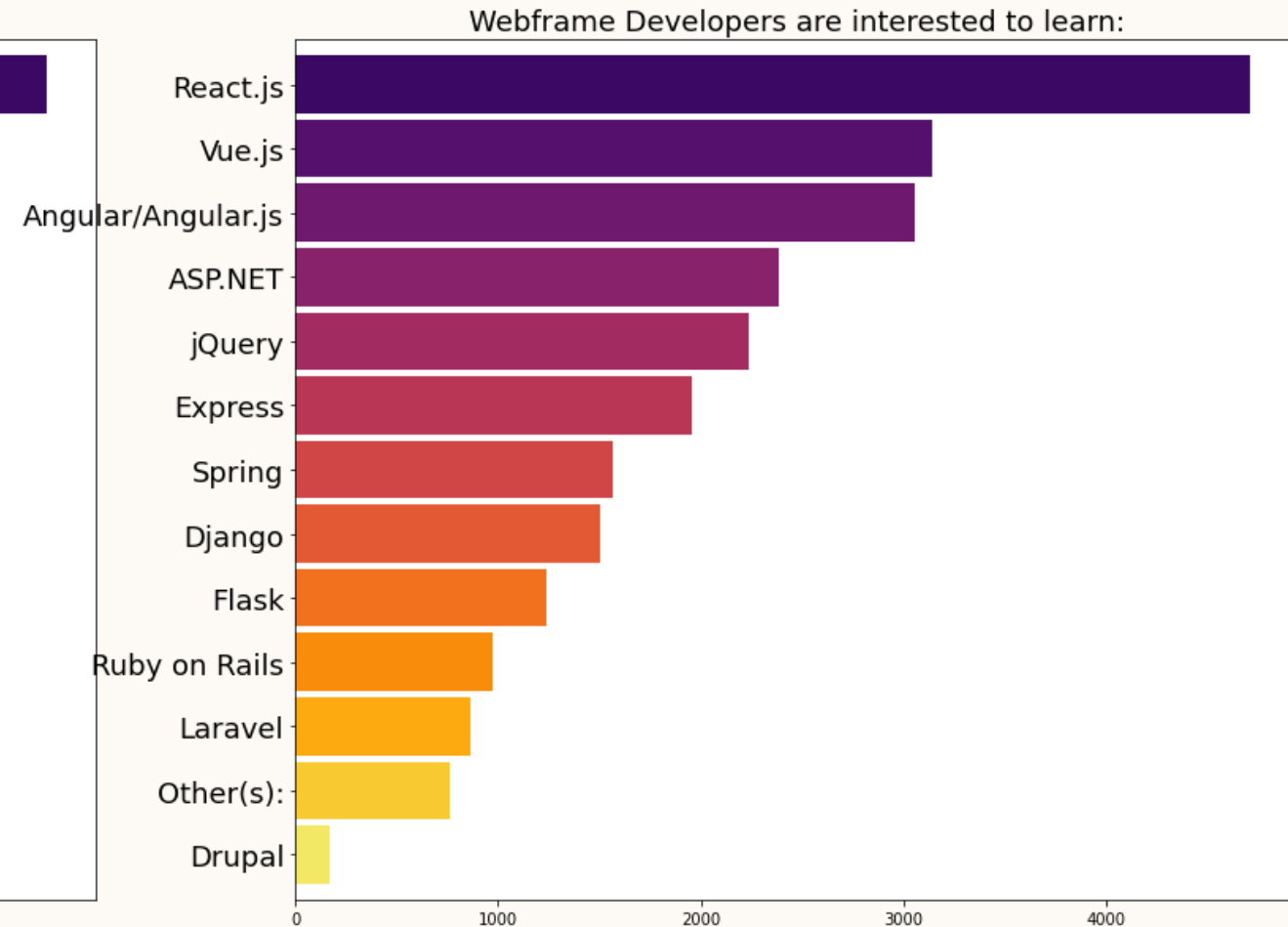
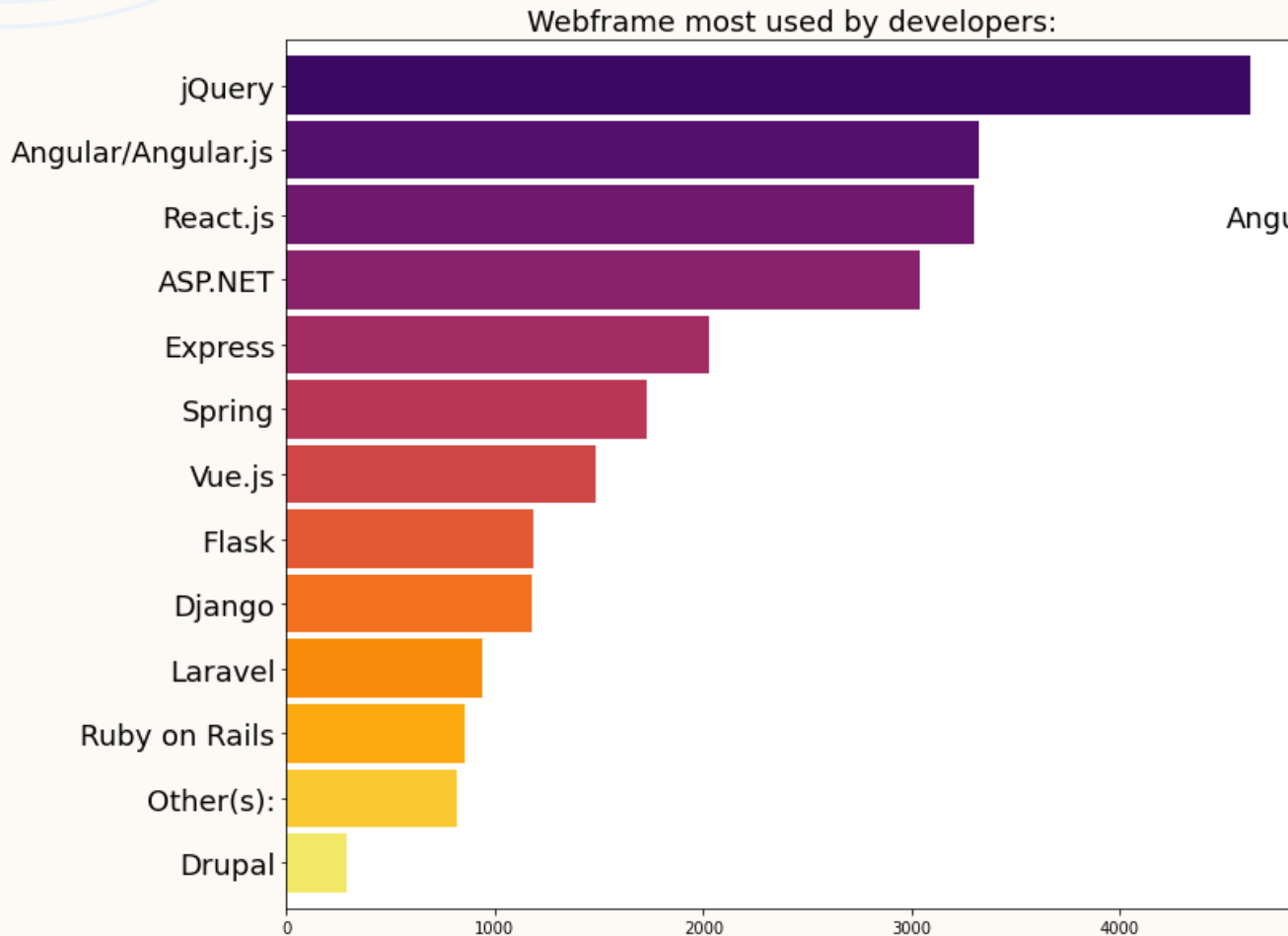


RESULTS & VISUALIZATIONS:

4. SOFTWARE USED

22

4. The Webframe most used can be seen as jQuery while the most they claim to want to learn is the popular React.js



In the normalized dataset, there were lots of missing data that could not be used. The below figure shows the percentage of data that is missing. The last 4 columns of data are 100 percent empty.

```
df_num_missing/len(df_lang)
```

Respondent	0.000013
LanguageWorkedWith	0.198391
LanguageDesireNextYear	0.252393
DatabaseWorkedWith	0.579367
DatabaseDesireNextYear	0.603499
PlatformWorkedWith	0.474353
PlatformDesireNextYear	0.405899
WebFrameWorkedWith	0.667476
WebFrameDesireNextYear	0.670639
DevEnviron	0.553479
OpSys	1.000000
Containers	1.000000
BlockchainOrg	1.000000
BlockchainIs	1.000000

dtype: float64

Other than that, we can say we have successfully conducted an analysis of the data.

The background features a large, light cream-colored circle on the left and a large, light pink circle on the right. These two circles overlap in the center. The top and bottom areas outside these circles are filled with a dark blue color. In the upper right portion of the pink circle, there are several thin, white, concentric curved lines that follow the circular shape.

THANK YOU